

NAEOS Whitepaper

Version 3.4.0

NAEOS Foundation

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NAEOS — Official Whitepaper

Nusantara Engineering & Architecture Operating System

“Specify Once. Build Anywhere.”

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Platform Version	v3.4.0 (repository state)

Executive Summary

NAEOS is an open-source declarative engineering platform that transforms specifications into high-quality software systems through a consistent, validated, and extensible pipeline. NAEOS is not just a project generator — it is an *engineering runtime* that understands specifications, builds an internal model (NEIR), orchestrates execution plans, generates artifacts, validates results, and keeps projects aligned with their specifications throughout the entire lifecycle.

Under the motto “**Specify Once. Build Anywhere.**”, NAEOS enables organizations to describe their system **once**, then derive code, documentation, configuration, and AI context from a shared engineering model across multiple languages and frameworks.

The current repository state includes Specification Language v2, a multi-adapter AI compiler, a NEIR-aware LSP, constitution-based governance, a marketplace, a production server daemon (**naeos serve**), SBOM and artifact signing, Helm chart scaffolding, and compliance-oriented policy templates for SOC 2, HIPAA, and GDPR workflows.

1. Background & Problem Statement

1.1 The Fragmentation of Software Engineering

The software industry faces a systemic fragmentation crisis:

- **Multi-language, multi-framework** — A single system now spans Go, TypeScript, Python, Java, and Rust simultaneously, each with its own frameworks, conventions, and toolchains.
- **Specification–implementation drift** — Documentation and code diverge over time; no automated mechanism guarantees alignment.

- **Loss of engineering context** — Architectural decisions, ADRs, and organizational knowledge live only in individuals' heads — undocumented and untraceable.
- **AI tooling explosion** — Each AI coding agent (GitHub Copilot, Claude Code, Cursor, Gemini CLI, Codex, OpenCode) uses a different context and instruction format, forcing organizations to maintain many configuration files with identical content but different formats.
- **Inconsistent governance** — Without enforcement mechanisms, policies, standards, and organizational rules remain static documents that are never executed.

1.2 The Cost of Inconsistency

The impact is directly measurable: rework, expensive audits, painful migrations, knowledge loss when team members leave, and failure to meet regulatory compliance (SOC 2, HIPAA, GDPR) due to a lack of verifiable audit evidence.

1.3 Thesis

The specification is the single source of truth. Everything — code, documentation, configuration, AI context, deployment artifacts — must be derived from the specification through a deterministic, validated, auditable pipeline.

NAEOS was built to prove this thesis.

2. Vision & Mission

Vision

Build an open-source engineering platform that enables developers and organizations to **describe their system once**, then build, validate, and evolve software across multiple languages, frameworks, and platforms — with an engineering constitution enforced automatically.

Mission

1. Make the declarative specification the single source of truth for every engineering artifact.
2. Provide a deterministic, validated, auditable compilation pipeline.
3. Bridge the gap between governance, specification, and execution through executable policies.
4. Build a universal bridge to all AI coding agents without vendor lock-in.
5. Deliver **technological sovereignty**: a vendor-neutral, language-neutral, cloud-neutral platform.

3. Foundational Principles: The Engineering Constitution

NAEOS formalizes its principles in the **Engineering Constitution** (NAEOS-CON-001) — the highest normative document, acting as the source of executable rules. Its normative hierarchy:



Figure 1: mermaid diagram 1

The twelve constitutional articles:

#	Article	Essence
I	Specification First	No specification = no implementation
II	Knowledge Preservation	Every engineering decision must be documented (ADR, RFC, API contract)
III	Traceability	Requirement → Spec → Architecture → Code → Test → Deployment
IV	Single Source of Truth	No two official artifacts may state conflicting normative information
V	Human Accountability	AI assists; humans decide and are accountable for releases
VI	Security by Design	Security is part of design from the start, not a final stage
VII	Documentation as Code	Documentation is versioned, reviewed, validated, and compiled
VIII	Reproducibility	Identical input produces identical output
IX	Vendor Neutrality	No dependency on a single AI vendor
X	Extensibility	Extension without modifying core specifications
XI	Quality Before Velocity	Speed must not sacrifice security, maintainability, correctness
XII	Continuous Improvement	The constitution evolves through RFC and ADR processes

What makes NAEOS distinctive: **the constitution is not merely a document** — its articles are compiled into executable rules enforced by the Rule Engine, executed by the Validator and Compiler, and checked by AI Review:

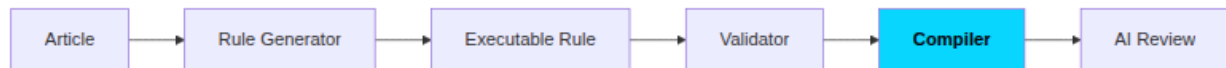


Figure 2: mermaid diagram 2

4. Platform Architecture

4.1 Layered Model

NAEOS connects five principal layers:

4.2 Compilation Pipeline

The core pipeline follows a deterministic flow:

The pipeline is backed by production-grade infrastructure:

- **Stage caching v2** — per-stage cache keyed by NEIR SHA-256 hash; hit rate inspectable via `--profile`.
- **Parallel generation** — concurrent multi-adapter execution ($\pm 1.4\text{ms}$ for 3 adapters vs $\pm 3\text{ms}$ sequential).
- **Profiling & memory profiler** — per-stage timing/memory, heap diffing, leak detection.
- **Pipeline middleware** — composable chain (log, metrics, auth, cache).
- **Event sourcing & observability** — execution snapshots, telemetry tracing, WebSocket live updates.

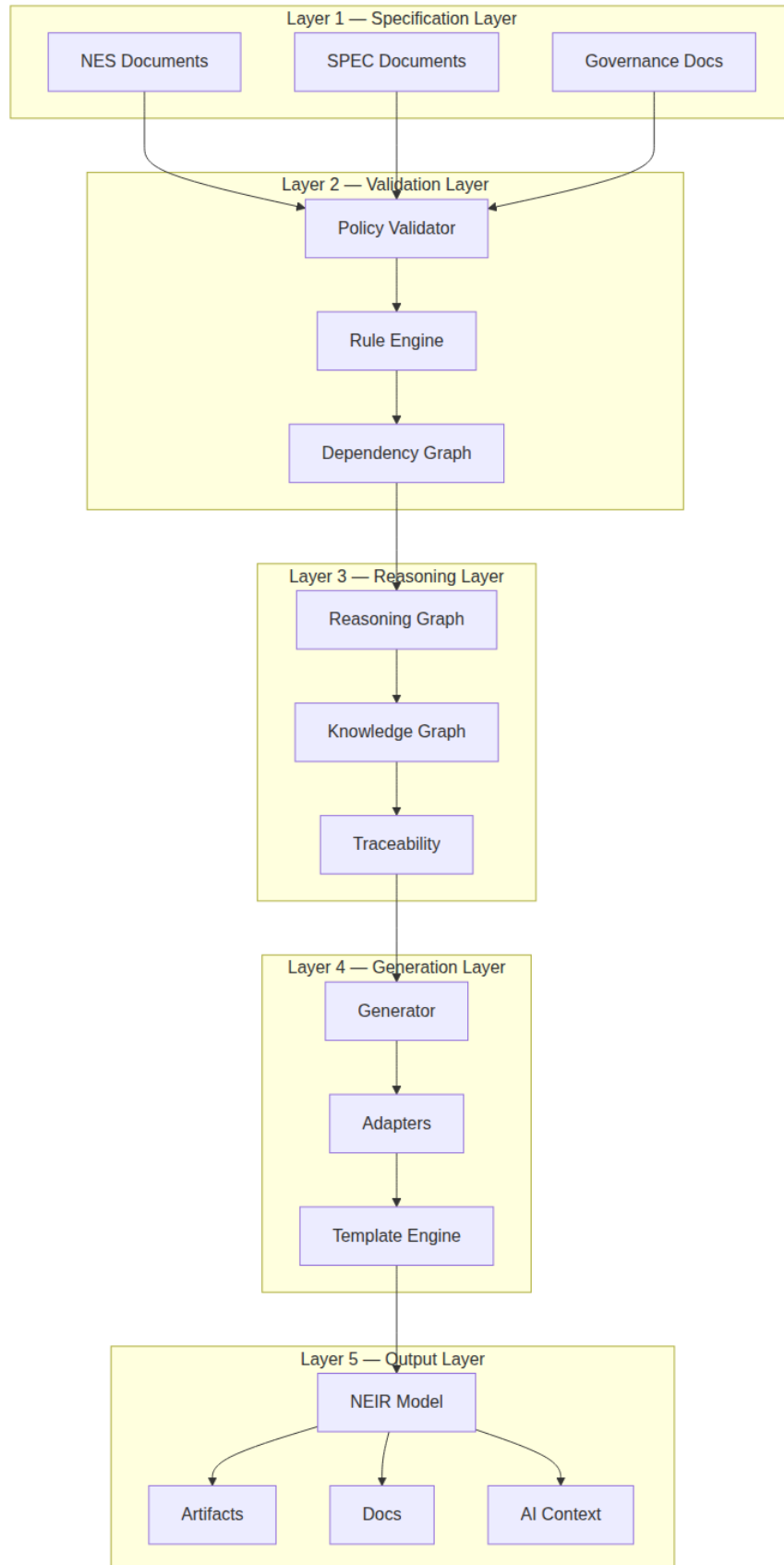


Figure 3: mermaid diagram 3

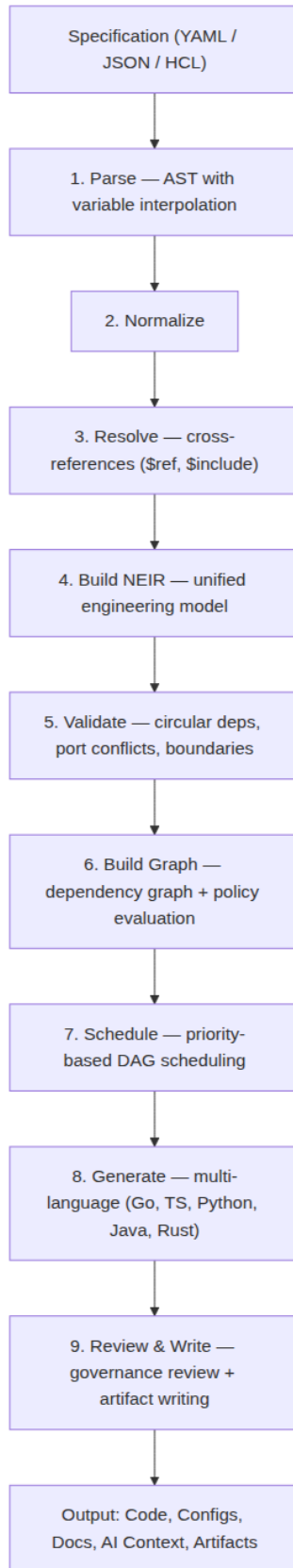


Figure 4: mermaid diagram 4

4.3 End-to-End Data Flow

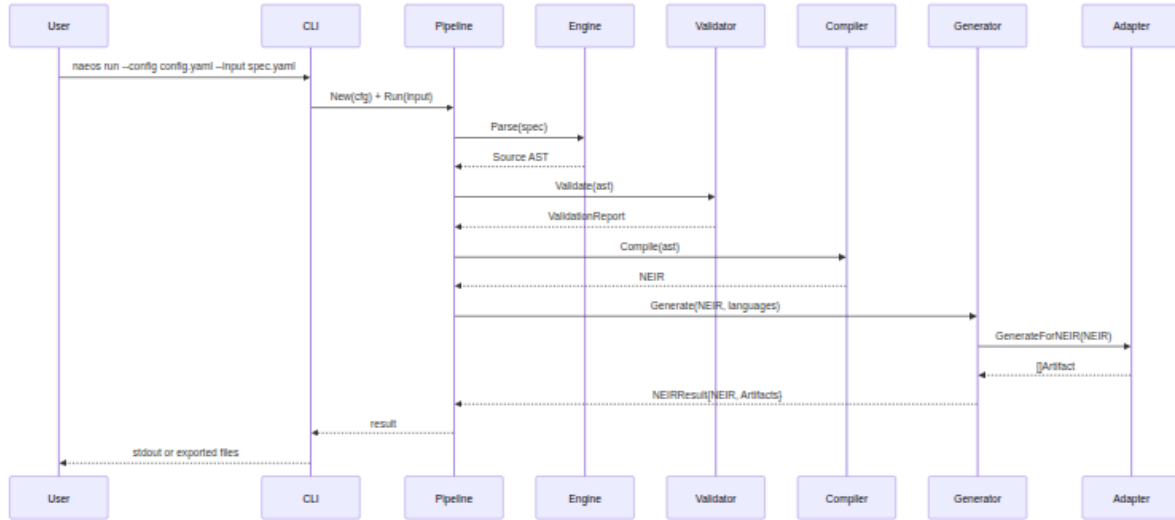


Figure 5: mermaid diagram 5

5. Core Components

5.1 Specification Language v2

The NAEOS specification language is the single source of truth — human-readable and machine-validatable:

Feature	Syntax	Purpose
Variable interpolation	<code>\${var}</code>	Reference values within a spec
Environment variables	<code>\$env{VAR}</code>	Resolve from environment
Cross-references	<code>\$ref{path}</code>	Reference between spec sections
File composition	<code>\$include{file}</code>	Multi-file specs
Built-in functions	<code>\$fn{upper/slug/default/...}</code>	Value transformations
Conditional sections	<code>\$if{cond}...\$endif</code>	Conditional content

Version 2.0 adds **conditional modules** (Condition field), **environment profiles** (ActiveProfile/Inherits), and schema-based validation with automatic minimum-version checking.

5.2 NEIR — NAEOS Engineering Intermediate Representation

NEIR is the **unified engineering model** representing the entire system. NEIR is not merely an AST — it is a canonical model spanning 14 domains:

project, architecture, domain, module, component, service, API, storage, infrastructure, security, AI, documentation, deployment, testing, metadata

Technical characteristics:

- **Lazy loading** — per-section accessors; only required data is loaded.
- **Schema versioning** — semver schema registry with remote validation (`naeos schema validate`).
- **Deterministic** — identical input produces identical NEIR (Article VIII).
- **Diff-able** — structural NEIR comparison to track system evolution.

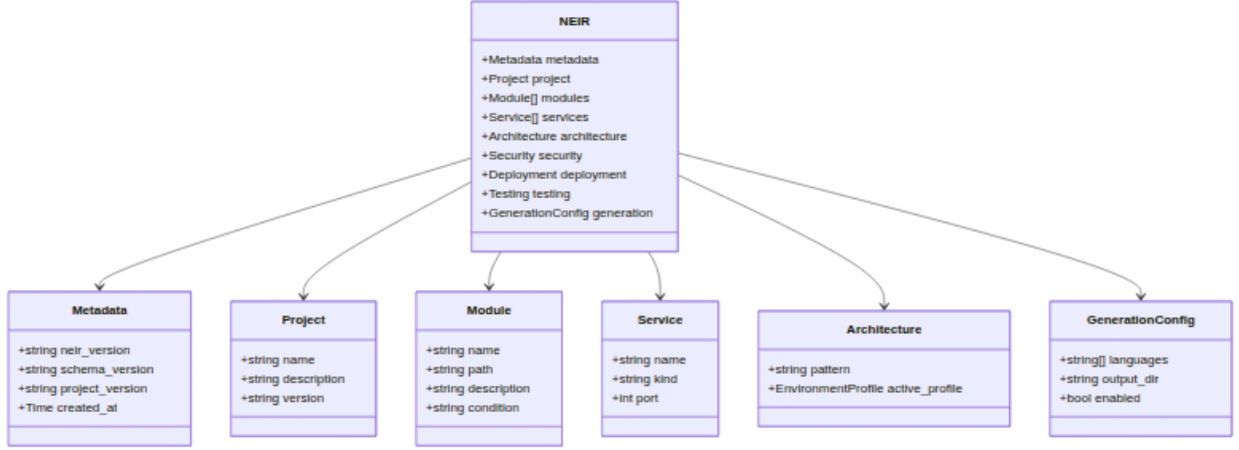


Figure 6: mermaid diagram 6

5.3 Multi-Language Generator

Language	Supported Stack	Status
Go	—	Active
TypeScript	—	Active
Python	—	Active
Java	JUnit 5	Active
Rust	Axum 0.7	Active

Each adapter produces consistent artifacts: code, configuration, Dockerfiles (5 languages), docker-compose, and Kubernetes manifests.

5.4 Kernel & Runtime

- **Service Registry** — centralized service registration
- **Event Bus** — internal pub/sub with PipelineObserver
- **Telemetry** — spans, batched export, HTTP exporter, Prometheus metrics
- **Lifecycle Management** — health checks, graceful shutdown, WebSocket draining

5.5 AI Integration & Compiler

The NAEOS compiler transforms NEIR into **AI instruction sets** for 6 target tools:

Additionally:

- **MCP Server** — Model Context Protocol for agent integration (validate_spec, compile_spec, list_artifacts, get_pipeline_status, export_terraform, list_plugins).
- **Context Bundles** — LLM-optimized project summaries enriched with dependency graphs, security context, and cloud resource mapping.
- **AI Compiler Adapter** — streaming spec compilation to LLMs (OpenAI, Anthropic, Ollama) with true SSE streaming.
- **Prompt Library** — centralized YAML-based prompt templates (LLM + compiler adapters) with custom template functions.
- **NEIR-aware LSP** — Language Server Protocol for spec YAML: autocomplete, diagnostics, hover, go-to-definition, code actions — with real parser integration.
- **VS Code extension** — extension generator (**naeos dx vscode-gen**) with TextMate grammar and LSP client.
- **AI Constitution** (NAEOS-CON-002) — articles specifically governing AI’s role in engineering, aligned with Articles V and IX.

5.6 Governance & Compliance

Policy & Governance: - Policy Evaluator — 7 operators, 5 built-in rules - Artifact Review — artifact inspection against governance rules - Audit Trail — traceable decision trail - Hierarchical RBAC — role-based access control

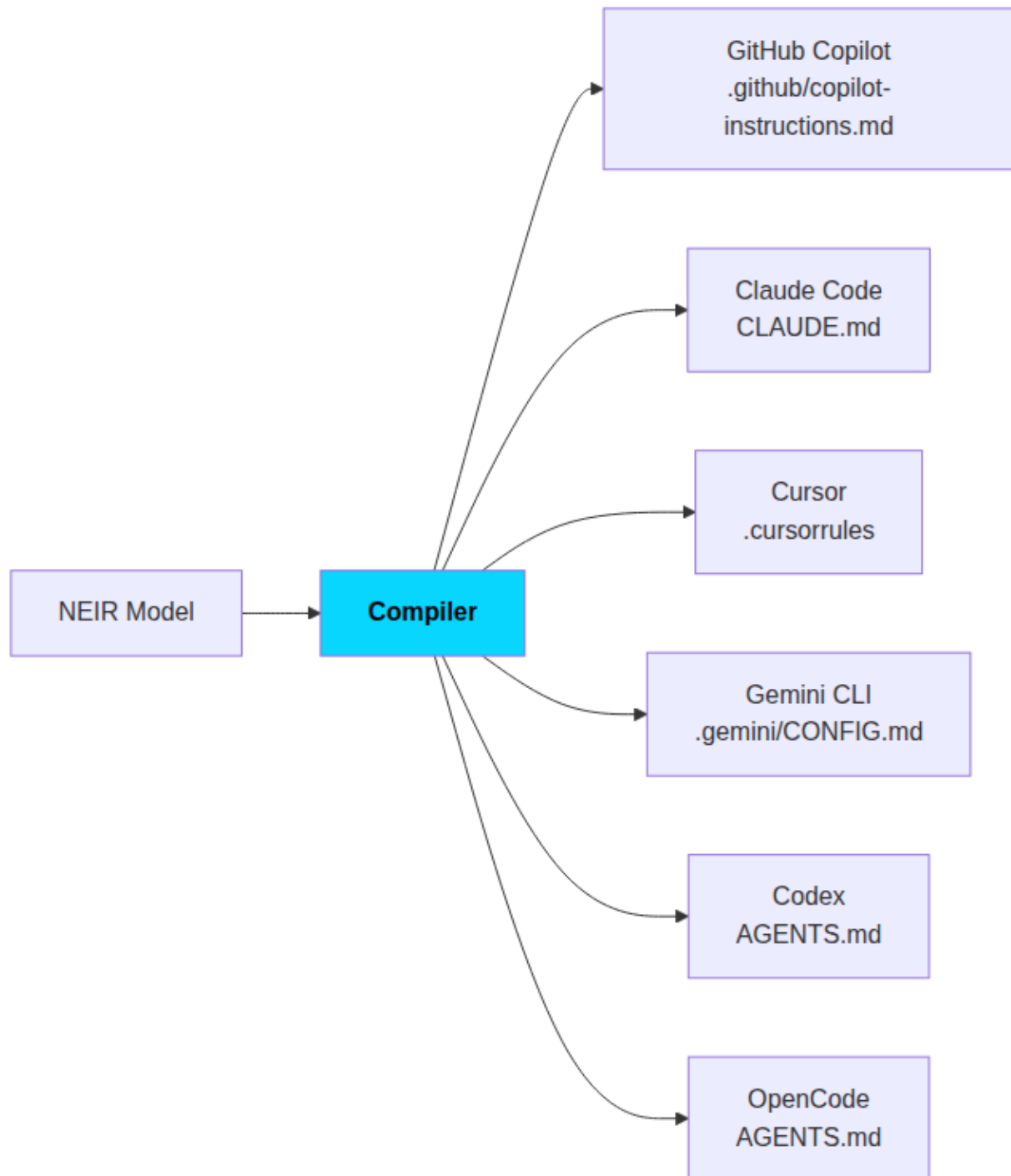


Figure 7: mermaid diagram 7

5.7 Ecosystem Marketplace

Marketplace	Function	Content
Profile	Publish, search, download	5 built-in industry profiles: SaaS, AI Agent, FinTech, Healthcare, Government
Plugin	Install/uninstall/search	WASM runtime (wazero), hot-reload, event bus, public registry
Template	Publish starter projects	Scaffolding with CI/CD, SDK, and WASM entry point

Plugins execute safely through a **JSON-over-stdin/stdout sandbox** and **WASI**, with signature verification and lazy loading.

6. Differentiation: NAEOS vs Conventional Approaches

Dimension	Conventional Approach	NAEOS
Source of truth	Many (code, docs, wiki, chat)	One: declarative specification
Code & spec	Drift over time	Derived together from spec, always aligned
AI context	Manual per-tool files, easily stale	Compiled from NEIR for 6 tools at once
Governance	Static documents, never executed	Constitution → Rule Engine → Validator, enforced automatically
Traceability	Manual, incomplete	Automatic: requirement → deployment
Compliance	Manual, expensive audits	Automatic reports (SOC 2/HIPAA/GDPR) + verifiable audit chain
Extensibility	Fork or separate tooling	Official plugin WASM, profile, template marketplace

7. Release History & Roadmap

Releases achieved

- **v0.x** — Foundation: parser, NEIR, pipeline, CLI, 6-adapter compiler, cloud (AWS/GCP/Azure), AI integration, distributed task execution, event sourcing
- **v1.x** — Stability: database layer (PostgreSQL/MySQL/SQLite), 999 lint issues resolved, production hardening, prompt library, observability dashboard
- **v2.x** — Platform: Supabase integration, NEIR v2.0 (conditional modules, env profiles), hierarchical RBAC, OAuth2/OIDC, SSO (SAML 2.0, LDAP), compliance frameworks, hashed + encrypted audit chains, stage caching, LSP, VS Code extension, real distributed builds, pipeline/memory profiling
- **v3.0.0** — Ecosystem release: 20+ new features, changelog, migration guide, deprecation notices
- **v3.1.0** — Performance release: pipeline caching on `naeos run`, run-level profiling (`--profile/--pprof`), architecture patterns (monolithic/microservices/serverless), WASM plugin hardening
- **v3.2.0** — Operations release: production server daemon (`naeos serve`) with TLS, graceful shutdown, and systemd integration; policy registry + control plane, runtime execution gateway, immutable evidence store, independent verification, MCP resources/prompts/completions/ping
- **v3.3.0** — Supply-chain release: SBOM generation (CycloneDX), Ed25519 artifact signing, SBOM verifier
- **v3.4.0** — Deployment release: Helm chart scaffolding, air-gapped bundles, config providers (env/file/K8s secret/Vault)

Platform health metrics (current)

Metric	Value
Test coverage	~87% (target 85%)
Lint pass rate	100% (linters, incl. gosec & errorlint)
CLI commands	200+ (284 CLI doc pages)
CLI test coverage	~80.8% (target 100%)
Packages 80% coverage	13+ (watch, rollback, cicd, distributed, gateway, websocket, configschema, monitoring, configreload, database, auth, supabase, and more)

Roadmap

- **v3.5.0** — Observability: OpenTelemetry (OTLP) tracing export, SLO & Prometheus alerting, audit export to SIEM
- **v3.6.0** — Scale: durable job queue (Postgres outbox), networked pipeline workers (NATS/Kafka), idempotency
- **v3.7.0+** — API v2, outbound webhooks, official SDKs, MFA/SCIM, per-tenant governance

See `DEVELOPMENT_PLAN.md` for the current project roadmap and milestone status.

8. Licensing & Project Governance

- **License:** Apache License 2.0 — free to use, modify, and distribute; commercial and internal use permitted with attribution.
- **Document governance:** all standards follow the NES (NAEOS Engineering Specification) flow, ADR (Architecture Decision Record), and RFC with mandatory review.
- **CI/CD:** every PR must pass lint + test + coverage checks; coverage drops block merges; every new API/feature must include documentation before merge.

- **Releases:** GoReleaser multi-platform builds (linux/darwin/windows × amd64/arm64) + multi-arch Docker image + automated blog post (EN/ID) per release.

9. Use Cases

Scenario	Value Delivered
Multi-language startup	One spec produces Go + TypeScript code and infra, removing ~70% of boilerplate
Regulated organizations (fintech/healthcare/gov)	Automated policy enforcement + SOC 2/HIPAA/GDPR reports + tamper-evident audit chain
Teams adopting AI coding agents	AI context compiled for 6 tools from one source — no more stale files
Multi-cloud enterprises	Terraform HCL for AWS/GCP/Azure from NEIR; industry profiles standardize architecture
Large evolving platforms	Structural NEIR diff, migration engine v0.1→v0.3, rollback, and spec repair
Teams enforcing quality	Comprehensive validation (circular deps, port conflicts), fuzz testing, standardized benchmarks

10. Risks & Adoption Considerations

- **Plugin ecosystem maturity** — Currently 0 community plugins; targets of 5+ (Q1 2027) and 20+ (Q3 2027). Mitigation: plugin SDK, template generator, and public registry are already available.
- **Spec language learning curve** — Mitigated by LSP, TUI wizard, and 57 NES specification documents.
- **AI determinism challenges** — Constitution Articles V and VIII ensure AI only assists within a deterministic pipeline; humans retain release decisions.

11. Conclusion

NAEOS offers a structural answer to modern software engineering fragmentation: a platform that enforces **the specification as the source of truth, the constitution as governing law, NEIR as the unified model, and AI as a curated partner** — not a replacement for human judgment.

Under Apache License 2.0, with a vendor-neutral architecture and a growing ecosystem, NAEOS invites organizations and communities to build a more disciplined, traceable, and reproducible future of engineering — where you describe your system **once**, and build it **anywhere**.

NAEOS Foundation — “Engineering With Discipline”

*This document is based on the actual state of the project (NAEOS-foundation/naeos repository, v3.1.0) and is intended for publication, technical evaluation, and adoption discussions. All technical claims can be verified in the official project documentation (docs/NES-, specification/, constitution/).**